PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Knowledge on blood donation and associated factors in Ethiopia: A systematic review and Meta-analysis
AUTHORS	Getie, Addisu; Wondmieneh, Adam; Bimerew, Melaku; Gedefaw, Getnet; Demis, Asmamaw

VERSION 1 – REVIEW

REVIEWER	MCCULLOUGH, JEFFREY University of Minnesota System
REVIEW RETURNED	05-Feb-2021

GENERAL COMMENTS	the authors are to be congratulated for undertaking this complex study in order to understand better the knowledge about blood donation in Ethiopia. The literature searching and statistics seem strong and well done. There is little discussion about these specific practices and kinds of information the authors focus on to determine whether there is adequate knowledge. Thus, the report would be strengthened by adding more descriptions of these kinds of knowledge and how they were graded or rated in order to determine the strength of information. The following are specific comments. Reference three is to a KAP study. I have not looked it up, but does it really verify that the large portion of the owners are replacement and family members? Next paragraph I did not see your reference 4 between numbers 3 and 5. Reference five seems to be another KAP study. Does it really contain data to support the statement that supply in developing countries is extremely low compared to the demand for blood? This would involve data showing availability and demand and reference 5 does not seem to indicate this. Page 5 line one HIDERS probably should be HINDERS. Another possible reference that could be used along with number 12 is Loenick at al which points out that concerns about blood donation are similar in many countries around the world. The database is searched, and the keywords and search process seem to have been very well done.

	Page 8 lines 2 - 4 wow did two investigators review at 1286 articles to identify the 20 used in this study. If so, this must've been a lot of work in these investigators are to be congratulated. In the analysis section the authors give four different percent values but three categories. Could the authors clarify this? In results it is impressive that 20 articles were identified describing the KAP of people in Ethiopia. This is quite impressive because I feel sure that this number of articles would not be found from the many other a much larger more populous country. Clearly there has been interest in this topic in this country. Next it is not clear from the method section how do a numerical value for prevalence was determined and what this means. Could the authors add an explanation? To determine a level of knowledge there must have been some knowledge criterion used and the responses to the criterion must have been graded somehow. While extensive statistics can be used to compare the results of different trials, within each trial the conclusion about the state of knowledge might be different from other trials. Possibly the statistical analysis accommodated for this and I might have missed it because I am not a statistician. Readers will want to know the kinds of information included in the articles that were reviewed and what the authors of those papers and if this paper considers good knowledge. Page 13 and table 4 con the authors explain a bit about what they mean by earlier in "influential" study? This term is based on the statistics but for readers many of whom will not be statisticians oh it will be helpful to explain what this means. Discussion section can the authors be clear about the meaning of the 56.57% knowledge about blood donation. Does this mean that is the portion of the population who knows about blood donation or does this number have to do with the extent of knowledge about blood donation? The authors are a tribute difference in the current study compared to the three other studies cited because of the potential difference in population however in the present study healthcare workers were included and I believe also university students so of those sub populations in the present study how do they relate to the results of the other studies focused on those particular groups of subjects? General it the authors compare the knowledge of blood donation in this meta- analysis with the knowledge of blood donation in other studies, but it is important for the readers of this manuscript to understand what is meant by knowledge of blood donation. Does this mean they have never heard of it and have no knowledge or that they know a little bit about it or a lot about it? If the latter, then the authors need to define what level of knowledge how the level of knowledge is determined and what level of knowledge is considered to be knowledgeable about blood donation. In the first paragraph page 16. I suggest the authors a look at the article by a Loenick et at and relate and use that to elaborate on the points made in this paragraph.

	Conclusion first sentence because the authors do not describe in this report what they consider adequate knowledge it is difficult to relate to the statement that knowledge about blood donation is it is not adequate. Lownik E, Riley E, Konstenius T, Riley W, McCullough J. Knowledge, attitudes, and practices of blood donation in developing countries: a systematic review of the literature. Vox Sang 2012; 103:64-74.

REVIEWER	Alvarez-Villalobos, Neri Universidad Autonoma de Nuevo Leon, Subdireccion de Investigación
REVIEW RETURNED	26-Feb-2021

GENERAL COMMENTS	It was a pleasure to read the research by Gertie Addisu et al. Project named "Knowledge on blood donation and associated factors among residents in Ethiopia: A systematic review and Meta-Analysis". This research, which in my opinion has many favorable points that contributes to the scientific community regarding the percentage of knowledge of blood transfusions and the factors associated with it. However, there are some important points that I would like to highlight to be modified and/or clarified: Starting with the title, it generates a bit of conflict for me to see the word residents in Ethiopia, since according to Dorland's Definition: resident: a graduate and licensed physician receiving training in a specialty, usually in a hospital. It makes me think that it could be a study in this population of doctors in training and not inhabitants of Ethiopia, this is not clear in the abstract or the introduction of the article. Upon reaching the materials and methods section and searching in the terms used for the search strategy, it was clear to me that a search was carried out to find: KNOWLEDGE in BLOOD DONATION and FACTORS ASSOCIATED IN PERSONNEL ASSOCIATED WITH THE SECTOR OF THE HEALTH. Understanding residents as Dorland`s definition. However, Gertie Addisu et al. make the comparison of illiterates against the above secondary school, which makes me think that it was in the general population (residents in the community), however, this search strategy could have limited the inclusion of articles that were not in health personnel. Probably to limit this problem I would add terms such as citizens, inhabitants, people, "residents in the community", "community residents", etc. Or simply remove the terms related to health providers. I know that there were 9 out of 20 articles included related to the community even when this term was not included in the search strategy. The background, the data sources, methods, limitations, and the study registration number were not included in the abstract of this study. The study eligibility criteria were not clear. Line 47 page 4 “The availability of blood for transfusion in a developing community is too limited due to the low mild lof knowledge, unwillingness to donate blood, and by blood donation practice”. Needs a reference. Line 47 page 4 author needs to include the % sign to 40.45. From the results section page 9 line 5, they mention that they reviewed 1286 articles and eliminated 571 due to duplication, 683 were excluded after title and summary, which makes a total of 32 and not 52, they need to review the numbers again. The PRISMA

	Flow diagram is correct except for the exclusion reasons that sum 31, and the authors need to clarify 32 reasons. The risk of bias and funding were not clearly included

VERSION 1 – AUTHOR RESPONSE

Reviewer 1 comments

The authors are to be congratulated for undertaking this complex study to understand better knowledge about blood donation in Ethiopia. The literature searching and statistics seem strong and well done. There is little discussion about these specific practices and kinds of information the authors focus on to determine whether there is adequate knowledge. Thus, the report would be strengthened by adding more descriptions of these kinds of knowledge and how they were graded or rated to determine the strength of information.

The following are specific comments.

Author response: We would like to say thank you for your encouraging and constructive comments. The authors of this manuscript accept the given comments and made necessary corrections accordingly. The answers to your comments and questions are incorporated and necessary amendments are done in the main manuscript as well as in the separated sheet as a point-by-point response as follows.

Reviewer comment 1: Reference three is on a KAP study. I have not looked it up, but does it verify that a large portion of the owners are replacements and family members?

Author response 1: Thank you; Reference three, which is cited as “Benedict N, Usimenahon A, Alexander NI, Isi A. Knowledge, attitude, and practice of voluntary blood donation among physicians in a tertiary health facility of a developing country. International Journal of Blood Transfusion and Immunohematology. 2012; 2 (4)” talks about KAP of voluntary blood donation and there is a sentence “In most developing and transitional countries, family replacement and paid blood donors are still a significant source of blood components for transfusion” in the introduction section of this reference, which indicates a large portion of the owners are replacements and family members rather than volunteer donors. However, to make it clearer, we amend the sentence in the introduction part under reference three as “Although adequate and safe blood is in high demand, there is a serious challenge of getting blood in stocks of developing countries. Blood donation practice is depending on the family replacement of remunerated blood donors. In most developing and transitional countries, family replacement and paid blood donors are a significant source of blood transfusion”.

Reviewer comment 2: Next paragraph I did not see your reference 4 between numbers 3 and 5.

Author response 2: Thank you for remembering this critical issue; reference 4 is missed which was a technical error and now included between numbers 3 and 5 after a sentence “The availability of blood for transfusion in Ethiopia is too limited due to the low level of knowledge, unwillingness to donate blood, and poor blood donation practice. This is a great challenge for patients who face blood loss due to different medical and surgical conditions (4)”. The idea of this sentence was taken from the conclusion section of reference “4” which was reported as “59.6% had poor knowledge and 52.3% had an unfavorable attitude towards blood donation”.

Reviewer comment 3: Reference five seems to be another KAP study. Does it contain data to support the statement that supply in developing countries is extremely low compared to the demand for blood? This would involve data showing availability and demand and reference 5 does not seem to indicate this.

Author response 3: Thank you; while we read and understand reference 5, it talks about the annual demand of blood for surgery and others like infectious disease like HIV/AIDS, which are more prevalent in low-income countries. However, the available blood is low as it says “Currently low-

income countries only receive 2.8 units, increasing the pressure on countries to source blood supplies unsafely". This has a synonymous meaning with a sentence "Even if every blood donor is considered as a hero, the amount of blood collected from blood donors and the average blood collection rate in developing countries including Ethiopia is low as compared to the demand for blood due to increased major operations and infectious diseases like HIV/AIDS (5)"
Reviewer comment 4: Page 5 line one HIDERS probably should be HINDERS.
Author response 4: Thank you; now it is corrected as "hinders"
Reviewer comment 5: Another possible reference that could be used along with number 12 is Loenick et al. which points out that concerns about blood donation are similar imany countries
Author response 5: Thank you for providing this reference; we include the reference "Lownik et al". along with number 12 and include the necessary information accordingly as "Even though the beliefs, attitudes, and behaviors regarding blood donation differ, more knowledge is needed among non-donors. Fear is the commonest deterrent to blood donation, whereas altruism is the commonest motivating factor to donate blood. There is a gap between willingness about blood donation and the number of donors in most populations around the world, which affect blood donation (13)".
Reviewer comment 6: The database is searched, and the keywords and search process seem to have been very well done.
Author response 6: Thank you for these encouraging and constructive comments.
Reviewer comment 7: Page 8 lines 2 - 4 wow did two investigators review 1286 articles to identify the 20 used in this study. If so, this must've been a lot of work which these investigators are to be congratulated.
Author response 7: Thank you for your appreciation; two authors (AG and AD) reviewed all articles, which takes a long period to identify the remaining 20 articles.
Reviewer comment 8: In the analysis section, the authors give four different percent values but three categories. Could the authors clarify this?
Author response 8: Thank you; it was good if you indicate the page number and the line where the percent value and the categories were found for making it easily accessible. From our understanding, in the analysis section, there are three percent values (25%, 50%, and 75%) and there are three categories (low, moderate, and high). If your concern is not matched with our answer, please indicate the page number and the line.
Reviewer comment 9: In the results, it is impressive that 20 articles were identified describing the KAP of people in Ethiopia. This is quite impressive because I feel sure that this number of articles would not be found in many other much larger, more populous countries. Clearly, there has been interested in this topic in this country.
Author response 9: Thank you for your encouraging comments.
Reviewer comment 10: Next, it is not clear from the method section how a numerical value for prevalence was determined and what this means. Could the authors add an explanation?
Author response 10: Thank you; we use a weighted inverse variance random-effects model at 95%CI to determine the numerical value for the pooled prevalence, which is found in the method section data processing and analysis sub-section.

Reviewer comment 11: To determine the level of knowledge, there must have been some knowledge criterion used and the responses to the criterion must have been graded somehow.
While extensive statistics can be used to compare the reults of different trials, within each trial the conclusion about the state of knowledge might be different from other trials. Possibly the statistical analysis accommodated for this and I might have missed it because I am not a statistician. Readers will want to know the kinds of information included in the articles that were reviewed and what the authors of those papers and if this paper considers good knowledge.

Author response 11: Thank you; we understand that different studies used different criteria to grade the level of knowledge. However, the studies included in our systematic review and meta-analysis used the mean score to determine the level of knowledge, and the analysis was done using a binary logistic regression model after categorizing into two categories (good knowledge and poor knowledge). Therefore, we calculate the pooled prevalence and the pooled odds ratio since they used a similar classification as presented in the outcome measurement section as “There are two main outcomes. These are the level of knowledge on blood donation and factors associated with the level of knowledge about blood donation. The level of knowledge was measured by using the mean score of knowledge questions. Thus, those study participants who scored mean and above on the knowledge assessment items were considered as knowledgeable, whereas those who scored below the mean score were considered as not knowledgeable about blood donation”.

Reviewer comment 12: Page 13 and table 4 con the authors explain a bit about what they mean by earlier in “influential” study? This term is based on statistics but for readers, many of whom will not be statisticians oh it will be helpful to explain what this means.

Author response 12: Thank you; to make it clear for all readers, we rewrite the sentence as “As revealed in Table 4, none of the point’s estimates outside of the overall 95% confidence interval, which confirms that the omission of any of the studies included in this systematic review and meta-analysis does not affect the overall prevalence of level of knowledge about blood donation”.

Reviewer comment 13: Discussion section can the authors be clear about the meaning of the 56.57% knowledge about blood donation. Does this mean that is the portion of the population who knows about blood donation or does this number have to do with the extent of knowledge about blood donation?

Author response 13: Thank you; our objective was to identify the pooled prevalence of knowledge in Ethiopia. Thus, 56.57% knowledgeable about blood donation indicates that 56.57% of the population had good knowledge (scored mean and above of knowledge assessment questions). For more clarification, we amend the sentence in the introduction section as “This systematic review and meta-analysis showed that 56.57% (95% CI: 50.30-62.84) of the population had good knowledge about blood donation”.

Reviewer comment 14: The authors are a tribute difference in the current study compared to the three other studies cited because of the potential difference in population however in the present study healthcare workers were included and I believe also university students so of those subpopulations in the present study how do they relate to the results of the other studies focused on those particular groups of subjects?

Author response 14: Thank you; in the previous three studies, the study participants in Malaysia, India, Nigeria, were nursing students, health care workers, and undergraduate medical students, respectively. However, in our study, the participants were all personnel in the community in addition to health care workers and university students. Therefore, the knowledge of blood donation might be higher among nursing students, health care workers, and undergraduate medical students than populations in Ethiopia (at the community level, university students, and health care workers) since we calculate the pooled prevalence. Generally, the knowledge about blood donation among pure nursing students, health care workers, and medical students usually higher than in the general population because of the presence of illiterates in the community even if others like health care workers and university students are there.

Reviewer comment 15: General it the authors compare the knowledge of blood donation in this meta-analysis with the knowledge of blood donation in other studies, but the readers of this manuscript need to understand what is meant by a knowledge of blood donation. Does this mean they have never heard of it and have no knowledge or that they know a little bit about it or a lot about it? If the

latter, then the authors need to define what level of knowing how the level of knowledge is determined and what level of knowledge is considered to be knowledgeable about blood donation.
Author response 15: Thank you; as per your recommendations, the authors discuss and make necessary amendments in the discussion section regarding what is the knowledge of blood donation, how to determine the level of knowledge, and what level of knowledge is considered knowledgeable about blood donation. However, the authors of this study tried to describe how the level of knowledge regarding blood donation and what level of knowledge is considered knowledgeable in the method section as “The level of knowledge was measured by using the mean score of knowledge questions. Thus, those study participants who scored mean and above on the knowledge assessment items were considered as knowledgeable, whereas those who scored below the mean score were considered as not knowledgeable about blood donation”.
Reviewer comment 16: In the first paragraph page 16. I suggest the authors look at the article by a Loenick et at and relate and use that to elaborate on the points made in this paragraph. Lownik E, Riley E, Konstenius T, Riley W, McCullough J. Knowledge, attitudes, and practices of blood donation in developing countries: a systematic review of the literature. Vox Sang 2012; 103:64-74 PubMed .(1)
Author response 16: Thank you for providing such relevant literature which may strengthen the discussion of this study. Thus, we incorporated the information found in the reference “Lownik E, Riley E, Konstenius T, Riley W, McCullough J. Knowledge, attitudes, and practices of blood donation in developing countries: a systematic review of the literature. Vox Sang 2012; 103:64-74 PubMed .” to elaborate on the points made in the first paragraph.
Reviewer comment 17: Conclusion first sentence because the authors do not describe in this report what they consider adequate knowledge it is difficult to relate to the statement that knowledge about blood donation is not adequate.
Author response 17: Thank you; we accept this comment and the conclusion is rewritten as “More than half of the study participants were knowledgeable about blood donation. Sex and educational status were significantly associated with the knowledge about blood donation in Ethiopia. Therefore, there is a need for education and dissemination of information among the general population about blood donation to build adequate knowledge and maintain a regular blood supply”

Reviewer 2 Comments
It was a pleasure to read the research by Getie Addisu et al. Project named "Knowledge on blood donation and associated factors among residents in Ethiopia: A systematic review and Meta-Analysis". This research, which in my opinion has many favorable points that contributes to the scientific community regarding the percentage of knowledge of blood transfusions and the factors associated with it. However, there are some important points that I would like to highlight to be modified and/or clarified:
Author response: We would like to say thank you for your encouraging and constructive comments. The authors of this manuscript accept all given comments and made necessary corrections accordingly. The answers to your comments and questions are incorporated and necessary amendments are done in the main manuscript as well as in the separated sheet as a point-by-point response as follows.
Reviewer comment 1: Starting with the title, it generates a bit of conflict for me to see the word residents in Ethiopia, since according to Dorland's Definition: resident: a graduate and licensed physician receiving training in a specialty, usually in a hospital. It makes me think that it could be a study in this population of doctors in training and not inhabitants of Ethiopia, this is not clear in the abstract or the introduction of the article. Upon reaching the materials and methods section and searching in the terms used for the search strategy, it was clear to me that a search was carried out to find: KNOWLEDGE in BLOOD DONATION and FACTORS ASSOCIATED IN PERSONNEL ASSOCIATED WITH THE SECTOR OF THE HEALTH. Understanding residents as Dorland`s definition. However, Getie Addisu et al. make the comparison of illiterates against the above secondary school, which makes me think that it was in the general population (residents in the community), however, this search strategy could have limited the inclusion of articles that were not in

health personnel. Probably to limit this problem I would add terms such as citizens, inhabitants, people, "residents in the community", "community residents", etc. Or simply remove the terms related to health providers. I know that there were 9 out of 20 articles included related to the community even when this term was not included in the search strategy.
Author response 1: Thank you; first we correct the title as “Knowledge on blood donation and associated factors in Ethiopia: A systematic review and Meta-analysis” to avoid confusion regarding the word “Residents” which was previously used to indicate the general population in Ethiopia. Then, as per your request, we kindly remove the term “related to health providers” since there is no change to the included articles
Reviewer comment 2: The background, the data sources, methods, limitations, and the study registration number were not included in the abstract of this study. The study eligibility criteria were not clear.
Author response 2: Thank you; the abstract is revised by considering your comments and by using PRISMA extension for abstracts (http://www.prisma-statement.org/Extensions/Abstracts.aspx) as “Objective: To assess the level of knowledge on blood donation and associated factors in Ethiopia.
Design: Systematic review and meta-analysis
Methods: Both published and unpublished cross-sectional studies conducted on knowledge of blood donation in Ethiopia were included. Articles reported other than the English language, qualitative studies, trials, case reports, review articles, updates, news, articles not reporting the outcome of the study, and articles without full text were excluded from the analysis. PubMed/MEDLINE, HINARI, EMBASE, Scopus, Google Scholar, and African Journals Online (AJOL) were searched for the study. Cochrane I^2 statistics were used to check the presence of heterogeneity. Subgroup analysis and sensitivity analysis were done on evidence of heterogeneity. Egger test with funnel plot was conducted to investigate publication bias.
Result: Twenty cross-sectional studies were included for the analysis of this systematic review and meta-analysis. A total of 8,338 study participants (4,712 males & 3,626 females) were involved. The overall pooled level of knowledge about blood donation was 56.57% (95% CI: 50.30-62.84). Having a secondary and above level of education (AOR=3.12; 95% CI; 2.34-4.16) and being male (AOR=1.81; 95% CI; 1.44-2.28) were factors associated with blood donation.
Conclusion: More than half of the study participants were knowledgeable about blood donation. Sex and educational status were significantly associated with knowledge about blood donation in Ethiopia. Therefore, there is a need for education and dissemination of information among the general population about blood donation to build adequate knowledge and maintain a regular blood supply”. We also rewrote the study eligibility criteria to make clear as “Both published and unpublished cross-sectional studies conducted on knowledge of blood donation among populations in the community, health care workers, and university students in Ethiopia were included whereas, qualitative studies, different trials, case reports, review articles, updates, and news were excluded from the analysis. Furthermore, articles not reporting the outcome of the study and those without full text were excluded”. However, this systematic review is not yet registered. As per the journal requirement regarding the abstract, it includes the following components (objectives, design, setting, participants, outcome measurement, result, and conclusion). Therefore, the abstract was written accordingly. The strength and limitations were presented next to the abstract “page 3” based on the requirements of the journal.
Reviewer comment 3: Line 47 page 4 “The availability of blood for transfusion in a developing community is too limited due to the low mild of knowledge, unwillingness to donate blood, and by blood donation practice”. Needs a reference.
Author response 3: Thank you; it was a technical error during the write-up of the manuscript. Therefore, we put a bibliographic reference number “4” at the end of the sentence “The availability of blood for transfusion in a developing community is too limited due to the low mild of knowledge, unwillingness to donate blood, and blood donation practice”.
Reviewer comment 4: Line 47 page 4 author needs to include the % sign to 40.45.
Author response 4: Thank you; sorry for this technical error, we corrected “40.45%” instead of 40.45.

Reviewer comment 5: From the results section page 9 line 5, they mention that they reviewed 1286 articles and eliminated 571 due to duplication, 683 were excluded after title and summary, which makes a total of 32 and not 52, they need to review the numbers again.
Author response 5: Thank you; it was a writing error, where 551 articles were excluded due to duplication rather than 571. Thus, as per your request, we revise the number again and necessary corrections were made as “The retrieved articles were 1286. Of these articles, 551 articles were excluded due to duplication. Six hundred eighty-three articles were further excluded after reviewing titles and abstracts. Of the remaining 52 articles, 32 were excluded which did not fulfill the inclusion criteria. Then, 20 articles were used for the final analysis”. For more clarification, you can check from the diagram (Figure 1, uploaded as an image file designation)
Reviewer comment 6: The PRISMA Flow diagram is correct except for the exclusion reasons that sum 31, and the authors need to clarify 32 reasons.
Author response 6: Thank you; we made the necessary corrections regarding the PRISMA flow diagram as “the outcome of interest not reported (n=30) and inaccessibility of the full text (n=2)”

Reviewer comment 7: The risk of bias and funding were not clearly included
Author response 7: Thank you; we amend the risk of bias as “The funnel plot showed that there is an asymmetrical distribution of studies included in the review and Egger’s test was statistically significant (p=0.02), suggesting the presence of publication bias “and regarding funding, since it is a systematic review and meta-analysis, funding is not applicable. Thus, we include the funding statement in the declaration section as “Not applicable”

VERSION 2 – REVIEW

REVIEWER	Alvarez-Villalobos, Neri Universidad Autonoma de Nuevo Leon, Subdireccion de Investigación
REVIEW RETURNED	29-Mar-2021

GENERAL COMMENTS	Due to the authors Getie et al. published in November 2020 a systematic review entitled: "Blood Donation Practice and Associated Factors in Ethiopia: A Systematic Review and Meta-analysis" in which their main objective was to assess the prevalence of blood donation practices and associated factors in Ethiopia, which is different from the current one in which the evaluation of the knowledge on blood donation is included, it makes me be a little more strict in considering this article suitable for publication, the main reason that the scientific contribution is limited, mainly because in the previous systematic review was carried out on the same dates(only 11 days apart) and it was practically the same search strategy, they only added the knowledge terms, and in the end both systematic reviews end up with the same 19 articles out of the 20 included. In this context I would like to see a publication that contributes to science and instead of seeing two regular articles, it would have been better to have only one good article. However, evaluating only the present article, I would like to thank the authors for the modifications they made to their article. In general, it seems to me that it could satisfy the requirements for publication, however, I still have some comments to remark: 1.- By having a previous systematic review with minimal differences regarding design, search strategy, included articles, analysis, research team, etc. It seems pertinent that an update of the search strategy should be carried out since the actual has

	more than 6 months, I would also suggest a modification to the strategy since the terms used it does not meet the adequate sensitivity to identify articles that solve the objective Research: "To assess the level of knowledge on blood donation and associated factors in Ethiopia." As this is not an article that seeks to identify knowledge only in health workers, the terms related to these should be eliminated from the search strategy ("university students" OR "residents" OR "health care workers" OR "health care providers" OR "health professionals" OR "collage of health science students") in this way, the search strategy will be sensitive enough to identify those articles that include evaluations of knowledge about blood transfusions, by citizens who are not workers in the health sector, in addition to those who work in the health sector. By updating the search strategy, we will be sure that in the past 7 months we are not losing new references that could enter the study, and when making the pertinent modifications to the search strategy (eliminate the terms "university students" OR "residents "OR" health care workers "OR" health care providers "OR" health professionals "OR" collage of health science students ") we will have greater confidence that we are not losing important articles that do not include this sector of health workers. Author Enawgaw is misspelled in the article (says Enawagaw), is well spelled in reference. As an additional comment, I would like to suggest including as bibliographic reference: Getie A, Wondmieneh A, Bimerew M, Gedefaw G, Demis A. Blood Donation Practice and Associated Factors in Ethiopia: A Systematic Review and Meta-analysis. Biomed Res Int. 2020 Nov 25; 2020: 8852342. doi: 10.1155 / 2020/8852342. PMID: 33294457; PMCID: PMC7714580. Although I am not using a program to identify plagiarism at the moment, it seems to me that using the previous systematic review as a template runs the risk of this problem, due to most of the bibliographic references are the same. Kind regards

VERSION 2 – AUTHOR RESPONSE

Reviewer 2 comments

We would like to say thank you for your encouraging and constructive comments. The authors of this manuscript accept the given comments and made necessary corrections accordingly. The answers to your comments and questions are incorporated and necessary amendments are done in the main manuscript as well as in the separated sheet as a point-by-point response as follows.

Reviewer comment 1: Due to the authors Getie et al. published in November 2020 a systematic review entitled: "Blood Donation Practice and Associated Factors in Ethiopia: A Systematic Review and Meta-analysis" in which their main objective was to assess the prevalence of blood donation practices and associated factors in Ethiopia, which is different from the current one in which the evaluation of the knowledge on blood donation is included, it makes me a little more strict in considering this article suitable for publication, the main reason that the scientific contribution is limited, mainly because in the previous systematic review was carried out on the same dates(only 11 days apart) and it was practically the same search strategy, they only added the knowledge terms, and in the end both systematic reviews end up with the same 19 articles out of the 20 included. In this context I would like to see a publication that contributes to science and instead of

seeing two regular articles, it would have been better to have only one good article. However, evaluating only the present article, I would like to thank the authors for the modifications they made to their article. In general, it seems to me that it could satisfy the requirements for publication, however, I still have some comments to remark:

Author response1: Thank you; we aimed to assess the level of knowledge on blood donation by reviewing different studies conducted in Ethiopia. The study Blood Donation Practice and Associated Factors in Ethiopia: A Systematic Review and Meta-analysis" in which their main objective was to assess the prevalence of blood donation practices and associated factors in Ethiopia" is done for a specific issue, which is blood donation practice differed from knowledge on blood donation. Thus, we separately reviewed articles conducted on the level of knowledge on blood donation in Ethiopia. Regarding the searching strategy, we conduct some amendment according to your comment as follow.

Reviewer comment 2: By having a previous systematic review with minimal differences regarding design, search strategy, included articles, analysis, research team, etc. It seems pertinent that an update of the search strategy should be carried out since the actual has more than 6 months,

I would also suggest a modification to the strategy since the terms used it does not meet the adequate sensitivity to identify articles that solve the objective Research: "To assess the level of knowledge on blood donation and associated factors in Ethiopia." As this is not an article that seeks to identify knowledge only in health workers, the terms related to these should be eliminated from the search strategy ("university students" OR "residents" OR "health care workers" OR "health care providers" OR "health professionals" OR "collage of health science students") in this way, the search strategy will be sensitive enough to identify those articles that include evaluations of knowledge about blood transfusions, by citizens who are not workers in the health sector, in addition to those who work in the health sector. By updating the search strategy, we will be sure that in the past 7 months we are not losing new references that could enter the study, and when making the pertinent modifications to the search strategy (eliminate the terms "university students" OR "residents "OR" health care workers "OR" health care providers "OR" health professionals "OR" collage of health science students ") we will have greater confidence that we are not losing important articles that do not include this sector of health workers.

Author response 2: Thank you again; as per your recommendation, we made different amendments to make a minimal difference regarding design, search strategy, included articles, analysis, research team, etc. as well as, to make search strategy sensitive enough, we eliminate the terms ("university students" OR "residents "OR" health care workers "OR" health care providers "OR" health professionals "OR" collage of health science students "). Then, the searching strategy is researched as (“knowledge” OR “awareness” AND “blood donation” OR “volunteer blood donation” AND “associated factors” OR “determinant factors” AND “Ethiopia”) and we found 335 articles from the database MEDLINE/ PubMed, 352 articles from Google Scholar database, and 7 from other databases, which gives a total of 694 articles searched from January 1st, 2000 up to May 01, 2021. However, only 20 articles fulfilled the inclusion criteria. Thus these 20 articles were used for the final analysis of this systematic review and meta-analysis.

Reviewer comment 3: Author Enawgaw is misspelled in the article (says Enawagaw), is well spelled in reference.

Author response 3: Thank you; it was a technical error and corrected as “Enawgaw”

Reviewer comment 4: As an additional comment, I would like to suggest including as bibliographic reference: Getie A, Wondmieneh A, Bimerew M, Gedefaw G, Demis A. Blood Donation Practice and Associated Factors in Ethiopia: A Systematic Review and Meta-analysis. Biomed Res Int. 2020 Nov 25; 2020: 8852342. doi: 10.1155 / 2020/8852342. PMID: 33294457; PMCID: PMC7714580.

Author response 4: Thank you for providing this reference. As per your suggestion, we provided the given bibliographic reference, which is cited as a reference number “19”.

Reviewer comment 5: Although I am not using a program to identify plagiarism at the moment, it seems to me that using the previous systematic review as a template runs the risk of this problem, due to most of the bibliographic references are the same.

Author response 5: Thank you; even if, the available articles are similar, we did not attempt to any plagiarism.

Reviewer comment 6: Competing interests of Reviewer: none declared Editor(s)' Comments to Author (if any)

Author response 6: Thank you

VERSION 3 – REVIEW

REVIEWER	Alvarez-Villalobos, Neri Universidad Autonoma de Nuevo Leon, Subdireccion de Investigación
REVIEW RETURNED	14-Jun-2021

GENERAL COMMENTS	It was a pleasure again to read the research by Gertie Addisu et al. Project named "Knowledge on blood donation and associated factors in Ethiopia: A systematic review and Meta-Analysis". Which, in my opinion, meets the necessary criteria for publication, due to its methodology and contribution to the Community.